

**TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING**

Khwopa College Of Engineering
Libali, Bhaktapur
Department of Computer Engineering



**A PROPOSAL ON
FACIAL RECOGNITION ATTENDANCE SYSTEM
USING OPENCV**

Submitted in partial fulfillment of the requirements for the degree

BACHELOR OF COMPUTER ENGINEERING

Submitted by

Gobinda Bista
Hari Krishna Khuju
Subash Bhandari
Suresh Shrestha

KCE/077/BCT/015
KCE/077/BCT/016
KCE/077/BCT/037
KCE/077/BCT/040

Under the Supervision of
Er.Dinesh Man Gothe
Department Of Computer Engineering

Khwopa College Of Engineering
Libali, Bhaktapur

**TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING**

Khwopa College Of Engineering
Libali, Bhaktapur
Department of Computer Engineering



**A PROPOSAL ON
FACIAL RECOGNITION ATTENDANCE SYSTEM
USING OPENCV**

Submitted in partial fulfillment of the requirements for the degree

BACHELOR OF COMPUTER ENGINEERING

Submitted by

Gobinda Bista
Hari Krishna Khuju
Subash Bhandari
Suresh Shrestha

KCE/077/BCT/015
KCE/077/BCT/016
KCE/077/BCT/037
KCE/077/BCT/040

Under the Supervision of
Er.Dinesh Man Gothe
Department Of Computer Engineering

Khwopa College Of Engineering
Libali, Bhaktapur

Abstract

The Facial Recognition Attendance System is an advanced solution designed to streamline and enhance attendance tracking processes. This system accurately identifies and records individuals' attendance in real-time. The technology's feasibility is evaluated across technical, economic, operational, legal, and ethical dimensions, highlighting its potential to revolutionize traditional attendance tracking methods. The traditional face recognition technology is more complicated for the extraction of facial image features and the selection of classifiers, and the recognition rate is not high. With the continuous maturity of the convolutional neural network from handwritten digit recognition to face recognition, A face recognition algorithm that tests CNN using the Python+OpenCV framework. This model can automatically extract facial image features and classify them. Using adam optimizer and Harr Cascade classifier for face recognition can make training faster convergence and more.

Contents

List of Symbols and Abbreviation	7
1 Introduction	1
1.1 Background Introduction	1
1.2 Problem Statement	1
2 Literature Review	3
3 Requirement Analysis	4
3.1 SOFTWARE REQUIREMENT	4
3.1.1 Python	4
3.1.2 OpenCV (Open Source Computer Vision Library)	4
3.1.3 MySQL	4
3.1.4 Tensor Flow	5
3.2 FUNCTIONAL REQUIREMENT	5
3.2.1 User Registration	5
3.2.2 Face Detection	5
3.2.3 Face Recognition	5
3.2.4 Enrollment and Database Management	5
3.2.5 Real-time Monitoring	5
3.3 NON-FUNCTIONAL REQUIREMENT	5
3.3.1 Performance	6
3.3.1.1 Response Time	6
3.3.1.2 Throughput	6
3.3.2 Scalability	6
3.3.3 Reliability	6
3.3.4 Availability	6
3.3.5 Security	6
3.3.5.1 Data Security	6
3.3.5.2 Access Control	6
3.4 FEASIBILITY STUDY	6
3.4.1 Technical Feasibility	7
3.4.1.1 Requirements Analysis	7
3.4.1.2 System Compatibility	7
3.4.1.3 Facial Recognition Technology	7
3.4.2 Economic Feasibility	7
3.4.2.1 Cost-Benefit Analysis	7
3.4.2.2 Return on Investment (ROI)	7
3.4.3 Operational Feasibility	7

3.4.3.1	User Acceptance	7
3.4.3.2	Training Requirements	7
3.4.3.3	Integration with Existing Processes	7
4	System (or Project) Design and Architecture	8
4.1	USE CASE DIAGRAM	8
4.2	System Block	9
4.3	Sequence Diagram	9
4.4	Class Diagram	10
5	Methodology	11
5.1	SOFTWARE DEVELOPMENT APPROACH	11
5.2	ALGORITHMS AND MODELS	11
5.2.1	Haar Cascades	11
5.2.2	Local Binary Patterns	12
5.2.3	Eigenfaces	12
5.2.4	Local Feature Analysis	12
5.2.5	Convolutional Neural Networks	12
5.3	Proposed System	12
5.3.1	Dataset Creation and Image Preprocessing	13
5.3.2	Face Detection	13
5.3.3	Face Recognition and Calculation	14
5.3.4	DataBase	14
5.4	Expected Outcomes	15
5.5	Conclusion	15

List of Figures

4.1	System Use Case Diagram	8
4.2	System Block Diagram	9
4.3	Sequence Diagram	9
4.4	Class Diagram	10
5.1	Prototype Model for Software Development	11
5.2	System architecture of face recognition system	13

List of Symbols and Abbreviation

CNN	Convolution Neural Network
DNN	Deep Neural Network
SVM	Support Vector Machine
MATLAB	Matrix Laboratory
SMPC	Shared Memory Parallel Computer
LDA	Linear Discriminant Analysis
RFID	Radio Frequency Identification
GUI	Graphical User Interface
PCA	Prompt Corrective Action
MLP	Multi-Layered Perception
HD	High Definition
HOG	Head Of Generation
LBP	Local Binary Patterns
ROI	Return on Investment
ROIs	Region Of Interface
SQL	Structured Query Language
LFA	Local Features Analysis
OpenCV	Open Source Computer Vision Library
RDBMS	Reational Database Managaement System
PHP	Hypertext Preprocssor

Chapter 1

Introduction

1.1 Background Introduction

Attendance tracking is a critical aspect of organizational management. Automated Attendance tracking has been applied to many industries as it offers more efficient and secure method for recording employee attendance. Traditional methods often are tedious, prone to errors and fails to handle large data. Hence, in recent years, there have been much interests in Automated Attendance tracking system. Therefore, many institutes started deploying many other techniques for recording attendance like use of Radio Frequency Identification (RFID) [3], iris recognition [4], fingerprint recognition, and so on. However, these systems are queue based which might consume more time and are intrusive in nature. Among them Facial Recognition has set an important biometric features. Face recognition technology is better than other biometric based recognition techniques like finger-print, palm-print, iris because of its non-contact process. Recognition techniques using face recognition can also recognize a person from a distance, without any contact or interaction with person. This proposal aims to implement a Facial Recognition Attendance System using OpenCV, a widely-used computer vision library, to enhance accuracy and efficiency in attendance management. The purpose of this system is to streamline and enhance our attendance tracking process.

1.2 Problem Statement

The current manual attendance tracking system is prone to errors, time-consuming, and lacks efficiency. Employees may forget to sign in, leading to inaccurate records. The manual process of collecting attendance data for all your employees will take time. You might also end up spending a lot of time maintaining scorecards, rectifying errors in time entry, and so on. Currently most of the facial recognition techniques is able to work fine only if the number of people in one frame is very few and under controlled illumination, proper position of faces and clear images. The proposed Facial Recognition Attendance System addresses these issues by providing

Accuracy: Facial recognition minimizes the risk of errors associated with manual data entry.

Efficiency: Real-time processing ensures quick and efficient attendance tracking.

Security: Facial recognition adds a layer of security, preventing unauthorized access and ensuring the integrity of attendance records

Chapter 2

Literature Review

In this paper [?], authors have designed and implemented an attendance system which uses iris biometrics. Initially, the attendees were asked to register their details along with their unique iris template. At the time of attendance, the system automatically took class attendance by capturing the eye image of each attendee, recognizing their iris, and searching for a match in the created database. The prototype was web based.

In this paper [?] “Face Recognition based Attendance System using Machine Learning Algorithms” by Radhika C. Damale, the author says identification of a person by facial features Known as facial recognition. A face feature can be used for various computer-based vision algorithms such as face recognition, emotion detection and multiple camera surveillance applications. Face recognition system is attracting scholars towards it. In this, different methods such as SVM, MLP and CNN are discussed. DNN is used to “face detection”. For SVM and MLP approaches, the features like PCA and LDA extracted using extraction algorithms. In CNN approach, images fed directly to CNN Module as a feature. The approach shows Good detection accuracy percentage for CNN based approaches. SVM, MLP and CNN achieve test accuracy of 87%, 86.5% and 98% on self-generated databases respectively.

In the paper [?] “Attendance System based on Face Recognition” written by Venkata Kalyan Polamarasetty, Muralidhar Reddy Reddem, Dheeraj Ravi, Mahith Sai Madala. they proposed that we need to catch the picture from the webcam or the outside camera.] To do as such, in MATLAB, they introduce the drivers from the math works site dependent on the sort of camera we are utilizing. Next, they use any rate 500 to 1000 catches of every individual. For getting higher level of exactness they used some HD camera so as to get results. For face distinguishing, we can do it utilizing the article falling class and we utilize the b-box technique. The caught countenances are trimmed into little pictures of goals 112x92. It would associate with 11 KB of size. The faces taken in the database are expected to stack into our workspace. We will stack the gallery images into that. All the HOG highlights separated are put away as exhibit list it restores a mark to which the given information matches or about coordinated.

Chapter 3

Requirement Analysis

3.1 SOFTWARE REQUIREMENT

Our Facial Regonition system requires Python, OpenCV, MySQL, TensorFlow, Flask, CNN, JavaScript, Keras which are described below;

3.1.1 Python

Python is a high-level interpreted programming language for general-purpose programming created by Guido van Rossum which as released in 1991. Python has design philosophy that emphasizes code readability, notably using significant white spaces. It provides constructs that enable clear programming in small and large scales. The programming language features dynamic type system and automatic memory management with support to multiple programming paradigms including Object Oriented Imperative, Functional and Procedural. The programming language has large comprehensive standard library.

3.1.2 OpenCV (Open Source Computer Vision Library)

OpenCV is a popular open-source library designed for computer vision tasks. It provides a wide range of functions and algorithms related to image and video processing. OpenCV is widely used for tasks like image and video processing, object detection, facial recognition, and more. It is written in C++ but has Python bindings, making it accessible to a broader audience.

3.1.3 MySQL

MySQL is an open-source relational database management system (RDBMS). Its name is a combination of "My", the name of co-founder Michael Widenius's daughter, and "SQL", the abbreviation for Structured Query Language. In our project, we use this database to feed into PHP system for data labeling. The database is created from scraping Nepali Sentiment text in python.

3.1.4 Tensor Flow

TensorFlow is an open-source deep learning framework developed by the Google Brain team. It is widely used for building and training machine learning and deep learning models. It provides Flexible architecture for building various machine learning models. It supports deep learning, including neural networks and convolutional neural networks (CNNs). TensorFlow is used for developing and training deep learning models for tasks such as image classification, object detection, natural language processing, and more.

3.2 FUNCTIONAL REQUIREMENT

3.2.1 User Registration

The system should allow administrators to register individuals by capturing and storing their facial features. Registration should be straightforward and user-friendly.

3.2.2 Face Detection

Accurate face detection algorithms should be implemented to identify and locate faces in images or video streams.

3.2.3 Face Recognition

Employ robust facial recognition algorithms for accurate identification of registered individuals. The system should provide a confidence score or percentage to indicate the level of certainty in the recognition process.

3.2.4 Enrollment and Database Management

The system should manage a secure database of enrolled users and their corresponding facial features. Allow easy addition, modification, or removal of users from the database.

3.2.5 Real-time Monitoring

Provide real-time monitoring capabilities for administrators to view attendance status. Generate alerts for late entries or other specified conditions.

3.3 NON-FUNCTIONAL REQUIREMENT

These requirements are not needed by the system but are essential for the better performance of sentiment engine. The points below focus on the non-functional requirement of the system.

3.3.1 Performance

3.3.1.1 Response Time

The system should have low latency in recognizing faces and recording attendance.

3.3.1.2 Throughput

It should be able to handle a specified number of facial recognition transactions per second.

3.3.2 Scalability

The system should be scalable to accommodate an increasing number of users and devices.

3.3.3 Reliability

The system should be highly reliable, minimizing the risk of false positives or negatives in facial recognition. It should have mechanisms to handle system failures and recover gracefully.

3.3.4 Availability

The system should have a high level of availability to ensure that it is accessible when needed.

3.3.5 Security

3.3.5.1 Data Security

Facial data and attendance records should be stored and transmitted securely, following industry best practices.

3.3.5.2 Access Control

Implement robust access control mechanisms to restrict unauthorized access to the system.

3.4 FEASIBILITY STUDY

A feasibility study for a Facial Attendance System is crucial to assess the viability of implementing such a system within an organization. The study typically examines various aspects to determine if the proposed solution is feasible from technical, economic, operational, and scheduling perspectives. Here are key components of a feasibility study for a Facial Attendance System:

3.4.1 Technical Feasibility

3.4.1.1 Requirements Analysis

Assess the technical requirements of implementing a Facial Attendance System, including the necessary hardware, software, and networking components.

3.4.1.2 System Compatibility

Evaluate the compatibility of the proposed system with existing infrastructure, databases, and devices.

3.4.1.3 Facial Recognition Technology

Investigate the reliability and accuracy of available facial recognition algorithms and technologies.

3.4.2 Economic Feasibility

3.4.2.1 Cost-Benefit Analysis

Evaluate the costs associated with developing, implementing, and maintaining the Facial Attendance System. Compare these costs with the expected benefits, including time savings, accuracy improvements, and potential cost reductions.

3.4.2.2 Return on Investment (ROI)

Calculate the expected ROI over a specified period, considering both tangible and intangible benefits.

3.4.3 Operational Feasibility

3.4.3.1 User Acceptance

Assess the willingness and ability of users (both administrators and employees) to adapt to and accept the new system.

3.4.3.2 Training Requirements

Identify the training needs for users and administrators to effectively use and manage the Facial Attendance System.

3.4.3.3 Integration with Existing Processes

Evaluate how seamlessly the system can integrate with existing attendance tracking and HR processes.

Chapter 4

System (or Project) Design and Architecture

We developed a Facial Recognition Attendance System which takes Face image as its input and process it to provide the value of matching face as its output. Then, it take decision of present and absent person.

4.1 USE CASE DIAGRAM

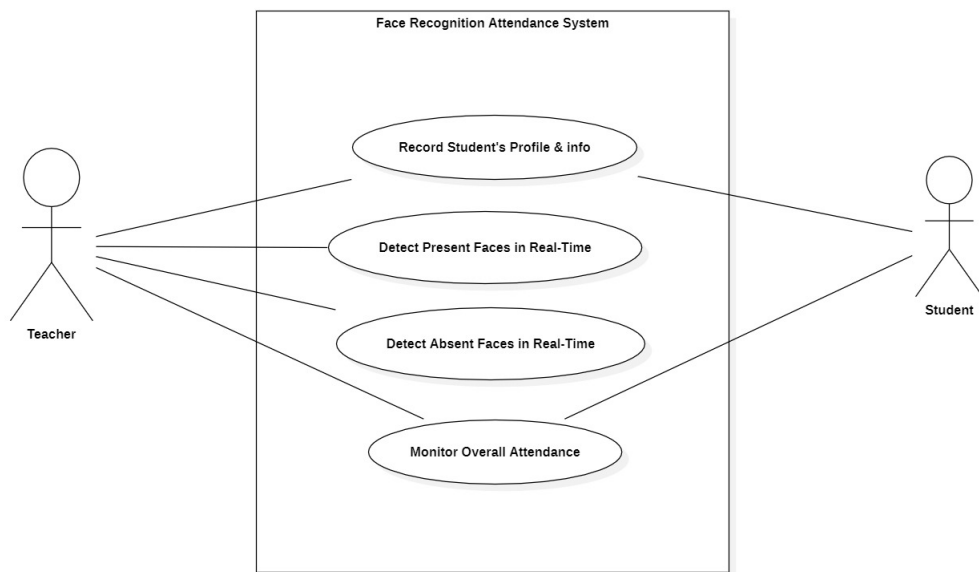


Figure 4.1: System Use Case Diagram

4.2 System Block

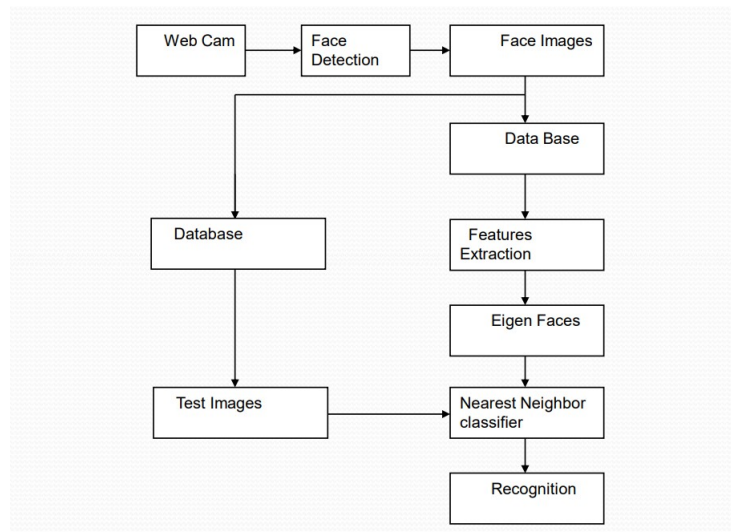


Figure 4.2: System Block Diagram

4.3 Sequence Diagram

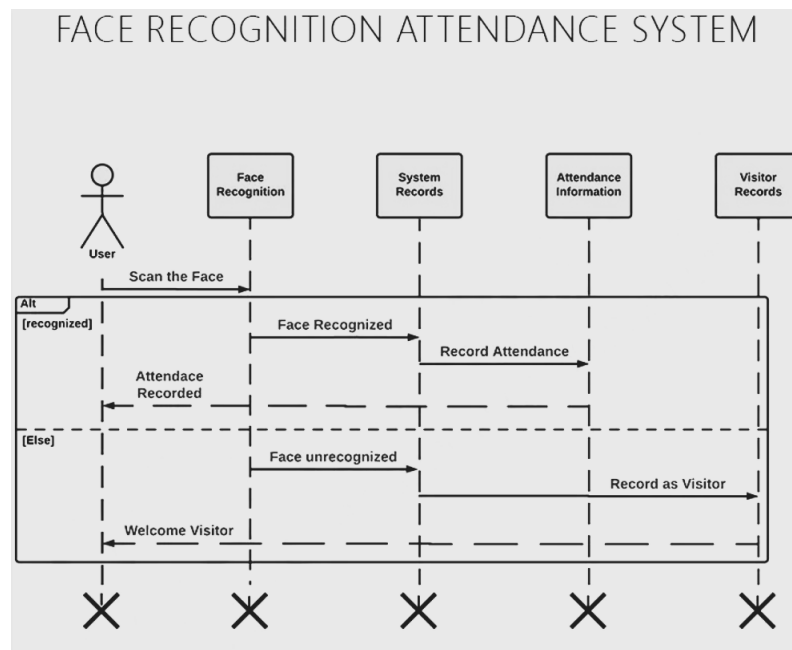


Figure 4.3: Sequence Diagram

4.4 Class Diagram

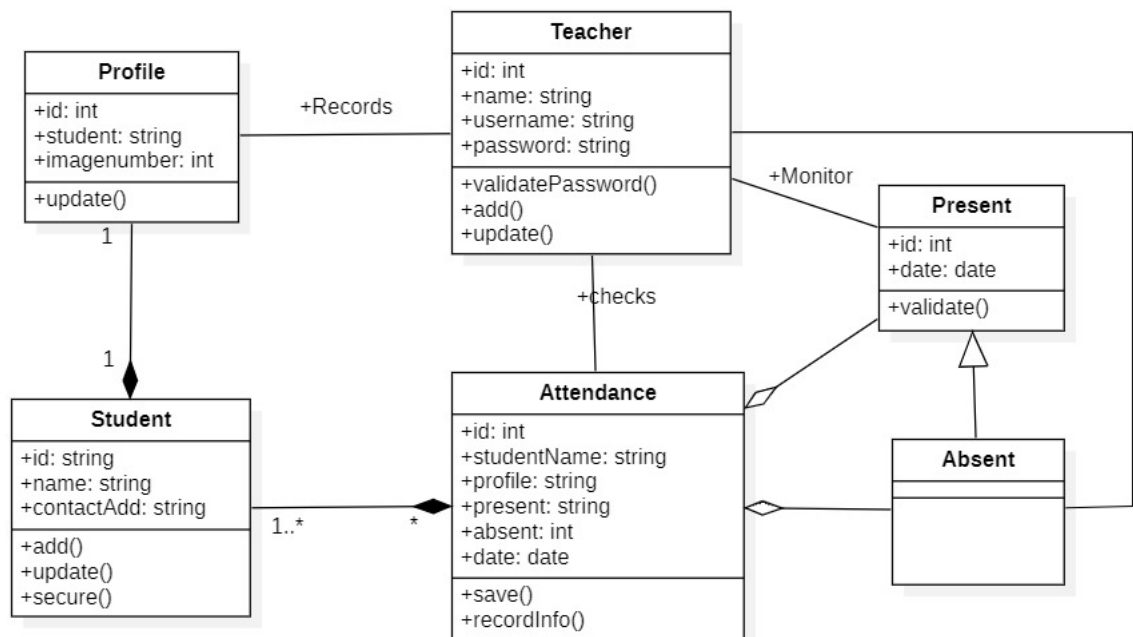


Figure 4.4: Class Diagram

Chapter 5

Methodology

5.1 SOFTWARE DEVELOPMENT APPROACH

Prototype model is a software development model where instead of freezing the requirements before design or coding can proceed, a throwaway prototype is built to understand the requirements. The prototype are usually not complete systems and many of the details are not built in the prototype. The goal is to provide a system with overall functionality. In this model, we create the prototype of the actual system, update the requirements and again rebuild the system until the final requirements are met.



Figure 5.1: Prototype Model for Software Development

5.2 ALGORITHMS AND MODELS

Facial recognition systems rely on various algorithms and models to accurately identify and verify individuals based on facial features. Here are some common algorithms and models used in facial recognition systems:

5.2.1 Haar Cascades

1. Description: Haar Cascades is an object detection algorithm used for identifying faces in images or video streams. It works by training on positive and

negative images to create a classifier.

2. Application: Initial face detection before more detailed analysis.

5.2.2 Local Binary Patterns

- Description: LBP is a texture descriptor that is often used for facial feature extraction. It focuses on local patterns of pixel intensities.
- Application: Feature extraction in facial recognition systems.

5.2.3 Eigenfaces

1. Description: Eigenfaces is a facial recognition algorithm that uses Principal Component Analysis to represent facial features in a lower-dimensional space
2. Application: Recognizing faces based on principal components.

5.2.4 Local Feature Analysis

1. Description: LFA focuses on extracting local features, considering facial regions independently. It enhances robustness against variations.
2. Application: Effective in handling facial variations due to expressions, pose, and lighting.

5.2.5 Convolutional Neural Networks

1. Description: CNNs are deep learning models that automatically learn hierarchical features from raw pixel data. They excel at capturing spatial hierarchies in images.
2. Application: State-of-the-art facial recognition, especially when dealing with large datasets and complex variations.

5.3 Proposed System

This system will be able to detect student or employee without physical contact and update the database based on the response from system. The system architecture of the proposed system is given below,

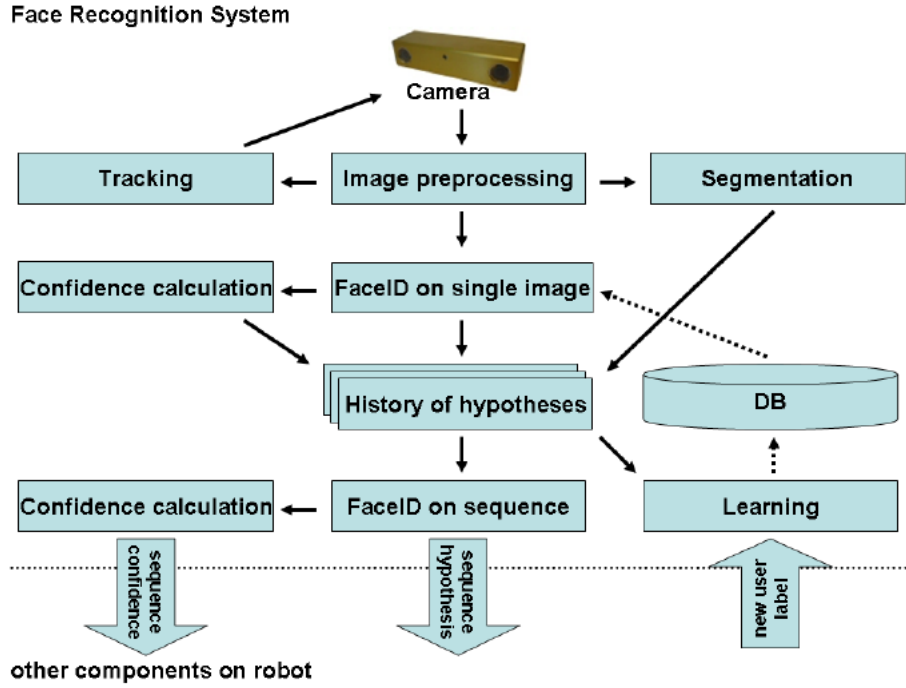


Figure 5.2: System architecture of face recognition system

5.3.1 Dataset Creation and Image Preprocessing

Images of Employee or student are captured using a web cam. Multiple images of single student will be acquired with varied gestures and angles. These images undergo pre-processing. The images are cropped to obtain the Region of Interest (ROI) which will be further used in recognition process. Then these images will be converted from RGB to gray scale images. This is done using OpenCV.

- Step1: we need to read the image with OpenCV's `imread()` function:
`img = cv2.imread(imagePath)`
This will load the image from the specified file path and return it in the form of a Numpy array.
`img.shape`
Output:
(1000,500,4)
- Step2: Convert the RGB data of Image into Greyscale To improve computational efficiency, we first need to convert this image to grayscale before performing face detection on it: Code
`gray_image = cv2.cvtColor(img, cv2.COLOR_BGR2_GRAY)` Output:
(1000,500)
This array only has two values since the image is grayscale and no longer has the third color channel.

5.3.2 Face Detection

The idea behind this technique involves using a cascade of classifiers to detect different features in an image. Haar Cascade algorithm needs to be trained to

detect human faces before it can be used for face detection. This is called feature extraction. The haar cascade training data used is an xml file- `haarcascade_frontalface_default`. We are using a file called `haarcascade_frontalface_default.xml`. This classifier is designed specifically for detecting frontal faces in visual input. `face_classifier = cv2.CascadeClassifier(cv2.data.haarcascades + "haarcascade_frontalface_default.xml")`. We are using `detectMultiScale` module from OpenCV. This is required to create a rectangle around the faces in an image. It has got three parameters to consider- `scaleFactor`, `minNeighbors`, `minSize`.

5.3.3 Face Recognition and Calculation

Face recognizer used in this system is Local Binary Pattern Histogram. Initially, the list of local binary patterns (LBP) of entire face is obtained. These LBPs are converted into decimal number and then histograms of all those decimal values are made. At the end, one histogram will be formed for each images in the training data. Later, during recognition process histogram of the face to be recognized is calculated and then compared with the already computed histograms and returns the best matched label associated .

5.3.4 DataBase

After face recognition process, Based on Calculated histogram Databases are updated .The name of employee or student along with day and time of attendance is also be stored in the database.

5.4 Expected Outcomes

The users can interact with the system using a GUI. This method will have higher accuracy in recognition of multiple faces from a single frame with lower response time.

S.N.	Model	Testing Accuracy
1.	Support Vector Machines (SVMs)	87%
2.	Multilayer Perception (MLP)	86.5%
3.	Convolutional Neural Network(CNN)	98%

The best model was found to be with Convolutional Neural Network(CNN) testing accuracy 98%. Other models performed worse than this model is due to, CNN functions automatically as a feature extractor, whereas ML algorithms function as a classifier. After building the basic CNN model we tried to increase its accuracy by combining 2D and 3D face recognition techniques, or using different machine learning algorithms to recognize faces more accurately. Other problem is to detect high variability of human faces in terms of shape, size, pose, expression, illumination, occlusion, and makeup. Haar Classifier technique can be adapted to accurately detect facial features. However, the area of the image being analyzed for a facial feature needs to be regionalized.

5.5 Conclusion

The implementation of a Facial Recognition Attendance System using OpenCV aligns with technological advancements, offering a more accurate and efficient solution for attendance tracking. This proposal lays the foundation for a robust system that addresses the current challenges in attendance management.